# Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

## 1 Cotton – Harvesting

### 1.1 Harvest Maturity Indices and Opening Stages

- **Harvest maturity indicators:** Harvest readiness in cotton (*Gossypium* spp.) is determined primarily by the proportion of open bolls across the canopy. Picking is initiated when bolls reach 60% to 70% open stage. Timely picking at this maturity threshold prevents open lint from being exposed to late-season morning dew, unseasonal rainfall, or falling honeydew and sooty mould.
- **Phenological timing and crop duration:** Total crop duration from sowing to final harvest spans 140 to 180 Days After Sowing (DAS), depending on cultivar duration, cropping zone, and irrigation setup. In hybrid and varietal seed production plots, cotton picking is specifically conducted between 140 and 150 DAS.
- **Visual and physical maturity signs:** Fully mature bolls crack along carpel sutures, turn brown and dry, and allow the lint to fluff out into a clean, white mass. Upon boll cracking, individual single-celled fibers undergo rapid desiccation, collapsing into flattened, kidney-bean cross-sectional ribbons with natural convolutions necessary for textile spinning.
- **Boll stratum selection:** In seed production fields, large-sized, fully mature bolls harvested strictly from the middle stratum of the plant canopy are selected to ensure high seed germination percentages and seedling vigor.

### 1.2 Pre-Harvest Cultural Practices and Chemical Defoliation

- **Irrigation termination schedule:** All irrigation operations must be completely terminated by the end of September in Northern canal-irrigated zones (or prior to final picking). Terminating irrigation at this stage hastens natural boll drying and uniform opening, prevents late-season boll rot, and stops unnecessary extension of the cropping cycle.
- **Chemical defoliation (Harvest Aids):** Foliar defoliants (such as ICAR-CICR defoliant formulation) are applied as a foliar spray when 60% to 70% of bolls are naturally open. Defoliants induce leaf abscission, drop mature foliage, improve sunlight penetration, and enhance canopy air circulation.
- **Lint quality protection via defoliation:** Removing foliage prior to harvest prevents trash, dry leaf bits, and green leaf stain contamination in picked seed cotton, thereby upgrading commercial lint classification and market value.

- **Deferral of late chemical sprays:** Chemical pesticide or fungicide applications should be deferred if the crop is naturally senescing and the majority of bolls have safely opened, preventing unnecessary input expenses and chemical residues on lint.

## 1.3 Picking Methods, Field Logistics, and Clean Harvesting Practices

- **Picking schedule and atmospheric conditions:** Seed cotton must be picked only when bolls are fully opened and completely dry during clear weather, after morning dew has evaporated. Harvesting wet cotton causes sunken discolored spots and promotes post-harvest mold development.
- **Manual picking mechanics:** Field labor manually picks seed cotton by twisting clean lint out of carpels rather than dragging or pulling stems, preventing mechanical damage to plant branches and avoiding dry carpel wall fragments in picked lint.
- **Segregation of picked cotton:** The first and last pickings are generally lower in fiber maturity and length; they must be harvested separately and never mixed with the main bulk produce. Clean bolls must be kept separate from stained, pest-infested, or partially rotted seed cotton.
- **Hybrid seed production picking protocol:** In GMS-based or hand-crossed hybrid seed production plots, pick bolls from male parent lines first and store them separately before harvesting crossed female bolls.
- **Contamination prevention guidelines:** Picked cotton must never be placed directly on wet soil, irrigation channels, or field bunds. Strict care must be taken to prevent synthetic contamination from Gutka/Pan Masala pouches, bidi/cigarette ends, candy wrappers, or plastic film, as non-lint trash severely degrades commercial lint classification.

## 1.4 Yield Potential, Ginning Outturn, and Fiber Quality Parameters

- **Harvested yield productivity:** Total seed cotton yield under irrigated, well-managed conditions averages 20 to 35 q/ha.
- **Individual boll yield components:** Average boll weight ranges between 3.5 and 6.0 g, containing 3 to 5 locules (locks) per boll.
- **Commercial fiber quality standards:** Marketable lint quality relies on high Ginning Outturn (GOT 32–40%), upper half mean fiber length (24–32 mm), bundle strength, and desirable micronaire value (3.5–4.9).
- **Seed quality specifications:** High-quality cottonseed harvested from middle-stratum bolls provides seed germination above 65–75%, providing clean, damage-free seeds suitable for acid delinting and subsequent sowing.

## 1.5 Post-Harvest Handling, Ginning Protocols, and Seed Cotton Storage

- **Storage hygiene and environment:** Harvested seed cotton must be stored in clean, dry, dust-free, damp-proof, and rat-free godowns or storage rooms.
- **Varietal and seed lot segregation:** Different cotton varieties, hybrids, and seed production lots must be stacked separately with clear tagging to maintain genetic and physical purity.
- **Ginning machine sanitation:** Ginning machinery (such as roller gins or saw gins) must be thoroughly cleaned between processing different cotton varieties to prevent mechanical seed mixing.
- **Post-harvest pest prevention:** Stored seed cotton piles must be covered properly with tarpaulin or heavy canvas to prevent post-harvest breeding and lint staining by dusky cotton bugs (*Oxycarenus hyalinipennis*) or diapausing pink bollworm larvae.

## 1.6 Ex-Situ and In-Situ Cotton Stalk Residue Management

- **Post-harvest livestock grazing:** Immediately after final picking, allow sheep, goats, or cattle to graze in the field. Grazing animals consume leftover unpicked bolls, damaged locules, and residual foliage, effectively destroying overwintering diapausing pink bollworm (*Pectinophora gossypiella*) larvae.
- **Ex-situ stalk removal and utilization:** Cut stalks at ground level and transport them away for domestic fuel (firewood) or industrial processing into fuel briquettes, brick kiln fuel, particle board, or bio-energy. Stalks must be stacked vertically away from cultivated fields and beaten against the ground to dislodge unopened or damaged bolls, which are subsequently burned or buried.
- **In-situ stalk shredding and incorporation:** Shred post-harvest stalks in the field using tractor-drawn multi-crop shredders, choppers, or mulchers, followed by incorporation into the soil with a rotavator.
- **Direct conservation sowing (Rabi crops):** Sow succeeding Rabi crops (such as wheat or mustard) directly into standing or shredded cotton residue using a Super Seeder or Happy Seeder, reducing field preparation costs and conserving soil moisture.
- **Biological decomposition enhancement:** Spray shredded stalk residue with *Trichoderma harzianum* or *T. viride* (@ 5 g/L water) before soil incorporation to accelerate organic decomposition and outcompete residual fungal root rot (*Rhizoctonia* spp.) and wilt pathogens.
- **Prevention of open field burning:** Open field burning of cotton stalks is strictly prohibited as it releases greenhouse gases, causes severe air pollution, destroys soil microbial fauna, and wastes valuable soil organic matter.

## 1.7 Off-Season Sanitation and Integrated Management Workflow

- **Off-season pest surveillance:** Install species-specific pheromone traps (@ 5 traps/ha) near stacked cotton stalks, village fuel piles, and local ginning mills to monitor off-season pink bollworm moth emergence.
- **Elimination of alternate weed hosts:** Clear perennial weed hosts (such as *Parthenium hysterophorus*) and volunteer/ratoon cotton plants from field bunds, water channels, and adjacent fallow lands during the off-season to destroy viral reservoirs (CLCuD, TSV) and deprive overwintering thrips and whiteflies of shelter.
- **Integrated step-by-step harvesting workflow:**
  1. *Pre-Harvest Stage (120–140 DAS):* Terminate irrigation by late September; inspect boll opening stage; apply foliar defoliants when 60–70% bolls are open.
  2. *Picking Operations (140–180 DAS):* Hand-pick clean, dry seed cotton after morning dew evaporates; segregate middle-canopy seed bolls and first/last pickings; prevent synthetic trash contamination.
  3. *Post-Harvest Field Clearance:* Allow livestock grazing; shred stalks in-situ with a Super Seeder or cut and stack ex-situ away from fields; treat residue with *Trichoderma*.
  4. *Storage and Off-Season Protection:* Store seed cotton in dry, rat-free godowns; gin varieties separately; install pheromone traps near stalk piles and ginneries to break pest carryover cycles.